

The Clear Lake Storm-Surge Markers: Historical and Technical Dossier for AR Reconstruction

Executive summary

The Clear Lake markers were real, unusually dramatic **predictive storm-surge markers** erected after Hurricane Ike as part of a joint federal/state public-awareness effort. Contemporary reporting describes a **\$56,000 federal-and-state project** intended to place large storm-surge markers in public locations along the upper Texas coast; later reporting based on a Texas Department of Public Safety spokesperson says the state and FEMA built **hundreds** of markers before the program was discontinued. Six markers were planned for the Houston portion of Clear Lake, but the strongest contemporary evidence identifies only **two confirmed installations**: one on **Clear Lake City Boulevard near Space Center Boulevard**, and one on **Bay Area Boulevard near Clear Lake High School**. ¹

The Houston installations were already controversial by an October 2010 town hall. Mayor Annise Parker halted installation of the remaining Clear Lake markers while the city reassessed their placement and possible “misinterpretation,” and City of Houston crews removed the two specifically identified markers around **mid-January 2011**. Residents and real-estate interests objected both to the implied flood risk and its potential effect on home values. Contemporary sources do **not** support the stronger story that FEMA was forced by Realtors to order the markers removed; the documented removal was done by Houston, after local complaints and lobbying of Councilmember Mike Sullivan. ²

The scientific controversy was more nuanced than “the marker was wrong.” The markers were labeled **Category 4** and **Category 5**, but just as they were being deployed, the National Hurricane Center was abandoning the practice of associating particular surge ranges with Saffir-Simpson categories. NHC explained in 2009 that category-specific surge ranges were “frequently incorrect” because surge also depends on storm size, track and landfall location; the modern Saffir-Simpson Hurricane Wind Scale therefore measures wind only. Councilmember Sullivan’s contemporaneous explanation that the marker language was inappropriate for the scales to be used “going forward” fits this national change extremely well, although I found no project document explicitly saying that the NHC change caused Houston’s decision. ³

Crucially, **that does not mean 20–25 feet of water in the Clear Lake/Galveston Bay system was scientifically implausible**. The contemporary *Bay Area Citizen* report says a state official characterized the marker’s 25-foot surge as a worst-case, approximately 500-year event. Later independent FEMA/USACE and Rice/SSPEED modeling examined extreme tracks capable of producing roughly **25 feet or more in parts of Clear Lake and still larger levels farther up the bay/Ship Channel system**. Those later models cannot be assumed to be the data source for the 2010 marker, but they undermine the idea that the pole was simply an absurd exaggeration. ⁴

The most important archival discovery for your AR project is the provenance of the photograph you remembered. Rice University’s Baker Institute now publishes a clear photograph as **Figure 3, “Flood**

Marker in Clear Lake Area Showing Surge Levels,” explicitly crediting the photo to Jim Blackburn himself. Blackburn had already written in a November 2018 Baker Institute issue brief that a Clear Lake Category 4/5 surge marker had been removed because it interfered with home sales. That fits your recollection of seeing Blackburn tell this story in roughly the 2018 period remarkably well, although I did not locate the exact slide deck or talk you attended. ⁵

The less satisfying finding is also important: I did **not** recover the original TDEM/FEMA project file, grant number, procurement package, fabrication drawing, manufacturer, exact installation coordinates, surveyed elevations, or documentation identifying the model/datum used to set the colored bands. Those should therefore be treated as unresolved rather than inferred. In particular, I would **not** yet build a “scientifically exact replica” by treating “25 feet” as 25 feet above local grade. The surviving reporting never establishes whether the marker was calibrated to depth above ground, water-surface elevation relative to a vertical datum, or a more generic public-education representation of a 25-foot regional storm surge. That distinction is critical in AR. ⁶

The research below follows the scope in your uploaded project brief. ^{filecite turn0file0}

Findings at a glance

Finding	Best current conclusion	Confidence
Sponsorship	Joint Texas/state + FEMA project; contemporary ABC specifically says FEMA funded the Clear Lake signs. ⁷	High
Total project cost	\$56,000 , described contemporaneously as federal/state funding. ⁸	High
Official program/grant name	Not recovered. No defensible HMGP/project number found.	Unknown
Geographic scope	Upper Texas coast; later DPS-sourced reporting says “hundreds” of markers were built for public locations. ⁹	Moderate-High
Clear Lake plan	Six planned in the Houston portion of Clear Lake. ¹⁰	High
Confirmed Houston installations	Clear Lake City Blvd near Space Center Blvd; Bay Area Blvd near Clear Lake High School. ¹⁰	High
Number actually installed in Clear Lake	At least two; <i>Chronicle/Bay Area Citizen</i> identifies two removed, while ABC says “several.” ²	Moderate
Appearance	Blue/green; Category 4 and Category 5 labels; tall blade-like/tapered body visible in Blackburn photo. ¹¹	High

Finding	Best current conclusion	Confidence
Physical height	Probably on the order of 20–25 ft; a later <i>Washington Post</i> account calls them 25-foot-tall signs, but no engineering drawing confirms this. ¹²	Moderate
Design surge	Contemporary reporting says marker illustrated water associated with a 25-ft storm surge ; ABC described water reaching as high as 20 ft in Clear Lake. ²	High that these figures were communicated; lower as engineering interpretation
Underlying model	Not identified. SLOSH/MOM is plausible but unproven.	Unknown
Removal	City of Houston crews removed confirmed signs around mid-Jan. 2011 after months of objections. ²	High
Did FEMA order removal?	No evidence found that it did.	Not established
Blackburn photo	Published by Baker Institute in 2023; caption says photo is by Blackburn. ¹³	High

Program origins, funding, and chronology

Hurricane Ike made landfall on the upper Texas coast on September 13, 2008. Ike demonstrated particularly starkly why wind category is a poor proxy for surge: it was a Category 2 hurricane at Texas landfall yet generated much more serious coastal inundation than “Category 2” sounded like to many members of the public. NHC’s contemporaneous advisory warned of coastal surge flooding potentially reaching **20 feet**, and in bays and rivers potentially **25 feet**, in the most exposed locations; subsequent NHC summaries describe roughly 15–20 feet above normal in portions of the Galveston Bay region. ¹⁴

The marker program followed Ike as a public-awareness measure. The February 1, 2011 *Bay Area Citizen/Houston Chronicle* report is the strongest contemporary account I found. It identifies the blue-and-green markers as part of a **\$56,000 “federal and state project” intended to raise coastal residents’ awareness of storm surge**. A 2016 ProPublica/Texas Tribune investigation, based in part on interviews with Texas officials, specifies FEMA and the state as the partners and describes the effort as building **hundreds of storm-surge markers for public locations on the upper Texas coast**. Texas DPS spokesperson Tom Vinger told the reporters the program was eventually “discontinued due to lack of community interest.” ¹

That is the limit of what I can substantiate about the funding mechanism. I searched indexed FEMA, TDEM/ Texas DPS, post-Ike disaster, hazard-mitigation and Houston sources but did **not** find an award record identifying a grant number or an official project title. FEMA disaster declaration DR-1791-TX covered Hurricane Ike, but the mere existence of DR-1791 does not establish that this \$56,000 education project was funded through a particular DR-1791 Public Assistance or Hazard Mitigation Grant Program award. It would be an overreach to call it an HMGP project without the missing grant file. FEMA’s current OpenFEMA HMA system does document project-level mitigation grants in general, while TDEM’s Hazard Mitigation Section

administers planning and mitigation grants in Texas, but neither public search produced the marker project itself. ¹⁵

Nor can the available \$56,000 figure be converted responsibly into a cost per marker. “Hundreds” is not an exact quantity, the sources do not establish whether \$56,000 included fabrication, installation, design, shipping, administration or all of those costs, and it is unclear how many fabricated markers were actually installed before local participation fell apart. ¹⁶

Chronology

Date	Event	Evidence and interpretation
September 13, 2008	Hurricane Ike strikes the upper Texas coast and demonstrates severe surge hazard despite Category 2 landfall winds.	NHC warned of extreme coastal/bay surge, including possible 20–25 ft conditions in exposed areas. ¹⁴
May 18, 2009	NWS/NHC proposes removing storm-surge ranges from the Saffir-Simpson scale.	NHC says category-specific surge ranges are often wrong and potentially misleading because surge depends on more than maximum wind. ¹⁷
2010, exact installation date unknown	Clear Lake markers are installed as part of the FEMA/state awareness effort.	Six were planned for the Houston portion of Clear Lake; at least two were erected. ¹⁰
By October 2010	Removal is already being discussed publicly.	At an October town hall, Councilmember Mike Sullivan said the city was working to have them removed because the wording was no longer appropriate. ¹⁰
2010, exact date unknown	Mayor Annise Parker halts additional Clear Lake installations.	Her letter says the city is “reassessing the markers and their locations due to possible misinterpretation.” ¹⁰
About mid-January 2011	Houston workers remove the two specifically identified installations.	Feb. 1 article says crews had removed them about two weeks earlier. ¹⁰
February 1, 2011	<i>Bay Area Citizen</i> publishes detailed removal story.	Provides \$56,000 figure, two locations, six planned markers, design description and political background. ¹⁰
February 9, 2011	ABC13 airs/reports the controversy.	Interviews Susan Maki, Todd Hinz and Mike Sullivan; explicitly says signs were FEMA-funded. ¹⁸
2016	ProPublica/Texas Tribune revisits the episode in its Houston hurricane-risk investigation.	Reports hundreds of markers, \$56,000 joint FEMA/state spending and DPS's explanation that the program ended for lack of community interest. ¹⁹

Date	Event	Evidence and interpretation
November 12, 2018	Jim Blackburn publishes the marker anecdote in a Baker Institute flood-policy brief.	He says the Clear Lake Category 4/5 sign was removed because it interfered with home sales. ²⁰
August 22, 2023	Blackburn and Jennifer Borski publish the photograph now most useful for reconstruction.	Baker Figure 3 credits Blackburn as photographer and identifies it as the Clear Lake surge marker. ¹³

The chronology exposes an important issue with the often-repeated “it lasted two months” formulation. Blackburn's 2023 retrospective says the marker was removed after two months, but Sullivan was already promising removal at an **October 2010** town hall, while the actual removal appears to have occurred around mid-January 2011. Therefore “two months” should be treated as an approximate later recollection, not used to calculate an installation date. ²¹

Clear Lake locations and coordinate reconstruction

The best contemporary location source is unusually specific about two installations but not specific enough to recover the exact pole bases. It states that Houston crews removed markers at **“Clear Lake City Boulevard, near Space Center Boulevard”** and **“Bay Area Boulevard near Clear Lake High School.”** It also says **six** markers had originally been planned for the Houston portion of Clear Lake. ¹⁰

Confirmed and planned sites

Site	Status	Best reference coordinate	Coordinate confidence	Evidence
Clear Lake City Blvd near Space Center Blvd	Confirmed installed and removed	≈29.5914, -95.1395	Low for the exact marker; moderate for the correct intersection vicinity	Contemporary report identifies the roads. The former Randall's/ Bay Pointe center at 2323 Clear Lake City Blvd is documented as being at the Clear Lake City/ Space Center corner; a geocoded business listing places the property at approximately 29.591385, -95.139511. This is a search/reference coordinate, not the recovered marker footing. ²²

Site	Status	Best reference coordinate	Coordinate confidence	Evidence
Bay Area Blvd near Clear Lake High School	Confirmed installed and removed	≈29.5821, -95.1056	Low for the exact marker; high that this is the correct school vicinity	Contemporary report says “near Clear Lake High School.” The school is at 2929 Bay Area Blvd; published coordinates are approximately 29.5821, -95.1056. Again, that is a school reference point, not a surveyed pole coordinate. ²³
Four additional Houston/ Clear Lake sites	Planned; apparently not completed, or at least not identified in recovered sources	Unknown	Unknown	Six were planned; the same contemporary account says Parker halted installation and names only the two markers Houston later removed. ¹⁰
Kemah	Considered but not installed under this initiative	N/A	High for non- installation as reported	Contemporary article says Kemah chose not to install markers amid concern that they would heighten fears. ¹⁰
Clear Lake Park	Not confirmed	N/A	—	I found no contemporary source placing one of these tall Category 4/5 markers in Clear Lake Park. The two documented Houston sites are roadside locations.

For your AR application, I would label the first two coordinates internally as something like `historicalSiteApproximation`, **not** `markerLocation`. Even an error of 30–100 meters will be visually obvious once you put a 20-plus-foot AR object against an intersection. The useful next archival target is therefore not more generalized web searching; it is an original **site list, work order, utility/installation ticket, construction photograph, or GIS layer** from TDEM/City of Houston.

There is also a clue in the Blackburn photograph that may eventually help geolocate the exact site. It shows the marker in a broad landscaped roadside setting with curb/road infrastructure and mature vegetation. Matching that historical scene against 2009–2011 Street View or archived aerial imagery may distinguish the Clear Lake City Boulevard installation from the high-school site. The Baker caption itself merely says Clear Lake area rather than naming which of the two sites is pictured. ¹³

Physical design and surviving imagery

The design is now much more constrained than it was from memory alone, but an engineering-grade reconstruction remains impossible without a second photograph or drawing.

The contemporaneous *Bay Area Citizen* description says the markers were **blue and green** and carried the words “**Category 4**” and “**Category 5**.” Blackburn's photograph confirms a tall, narrow two-color marker: the lower/main portion is blue and the upper portion green, with large white vertically arranged category lettering/numerals. A small rectangular light-colored information panel or headpiece appears near the top, and a narrow structural element continues above/behind it. ¹¹



Rice University Baker Institute, Figure 3. The publication identifies this as the Clear Lake flood marker and credits the photograph to Jim Blackburn. The capture date is not supplied. ¹³

Direct image:

<https://www.bakerinstitute.org/sites/default/files/2023-08/Blackburn-6years-Harvey-080423-Fig03.png>

Baker source page:

<https://www.bakerinstitute.org/research/assessing-houstons-flood-vulnerability-6-years-after-harvey>

Reconstruction specification from evidence presently available

Feature	What the evidence supports	Confidence / AR treatment
Overall form	Very tall, narrow, freestanding vertical marker with a distinctive shaped/tapered panel rather than a conventional rectangular road sign. The photographic silhouette broadens through portions of the body and narrows toward an end. ¹³	High visually , but trace the photograph rather than inventing a symmetrical profile.
Colors	Lower/main field blue , upper field green , white lettering. ¹¹	High
Labels	"CATEGORY 4" and "CATEGORY 5," prominently displayed. ¹¹	High
Height represented	Article says the marker was designed to show how high water from a 25-foot storm surge could get. ABC describes a sign showing water could reach as high as 20 ft in Clear Lake. ²	High that 20–25 ft was the public message; do not yet equate this to exact object height.
Physical overall height	Later <i>Washington Post</i> reporting describes the Clear Lake signs as roughly 25 feet tall . No original specification recovered. ¹²	Moderate. A 25-ft provisional AR model is reasonable if labeled reconstructed/ approximate.
One-foot graduations	No clearly documented one-foot graduation specification was recovered.	Unknown. Do not add one-foot ticks merely from memory to a "historically accurate" version.
Alternating black/white sections	The definitive Blackburn photo instead shows a blue/green body. ¹³	Not supported for the main marker body. A separate staff/ component may explain the recollection, but evidence is insufficient.
Top placard wording	A top/head element is visible, but its small text cannot be read reliably in the available reproduction. ¹³	Unknown. Leave blurred/blank in a high-integrity reconstruction until another source is found.
Width / exact profile dimensions	No dimensions recovered.	Unknown. Derive provisional proportions from photo pixels rather than claiming dimensions.

Feature	What the evidence supports	Confidence / AR treatment
Material	No procurement/specification source recovered.	Unknown. The photo alone is insufficient to distinguish aluminum, composite signboard, painted sheet, etc.
Support/foundation	Freestanding roadside installation is evident; exact structural attachment is not documented. ¹³	Partial visual evidence only.
Manufacturer/fabricator	Not recovered.	Unknown
Graphic designer	Not recovered.	Unknown
Engineering/shop drawings	Not recovered from indexed public sources.	Highest-priority archival target.

One potential source of confusion is that contemporary reporting discusses a **25-foot surge**, while a later account describes **25-foot-tall signs**. Those might both be correct, but they are not logically equivalent. Until the construction drawing appears, the defensible language is: **“The marker represented approximately 20–25 feet of potential water; later reporting describes the sign itself as about 25 feet tall.”** ²⁴

Recoverable image and video trail

The Blackburn/Baker image is by far the best authenticated photograph I recovered. The 2023 article gives it unusually strong provenance: the caption explicitly says **“Source Photo by the author,”** and Blackburn is an author of the report. It does not give the date on which he photographed the marker or identify which of the two known installations appears in the frame. ¹³

The two contemporary media pages are also valuable archival leads:

Media item	What survives publicly	AR/reconstruction value
Mary Ann Hellinghausen, “Storm surge markers removed,” Feb. 1, 2011	Article survives and contains a photo slot in the current page structure; the indexed text gives excellent physical/location description, but I did not recover a separately addressable original-resolution marker photograph from it. ¹⁰	Very high archival target because the newspaper's photo archive may contain contemporaneous views.
ABC13/KTRK, “Clear Lake removes storm surge warning signs,” Feb. 9, 2011	Archived article survives with interview transcript-like quotations. Current searchable version does not expose original B-roll/stills to this research interface. ¹⁸	Extremely high target: the broadcast package may show the sign from several angles and could contain people/cars useful for scale.

Media item	What survives publicly	AR/reconstruction value
Jim Blackburn/Baker Institute Figure 3, published 2023	Clear, authenticated single photograph; original source is Blackburn. ¹³	Best currently accessible reconstruction source.

ABC archive page:

<https://abc13.com/archive/7946821/>

Contemporary *Chronicle/Bay Area Citizen* page:

<https://www.chron.com/neighborhood/bayarea/news/article/Storm-surge-markers-removed-9299639.php>

A particularly worthwhile records request to **KTRK/ABC13** would cite the exact air/publication date, February 9, 2011, and title. News stations often retain raw packages or library footage even when an old web video player disappears.

Surge science, model provenance, and vertical-datum problem

What we know about the marker's science

The contemporary source makes one unusually important technical statement: the Category 4/5 marker was designed to show water associated with a **25-foot storm surge**, and an unnamed state official characterized that as a **worst-case scenario associated with a 500-year event**. That is currently the closest thing to a description of the design basis. ¹⁰

What the source does **not** tell us is equally important. It does not name SLOSH, HAZUS, a FEMA Flood Insurance Study, a hurricane evacuation study, an ADCIRC simulation, a particular synthetic storm, a tide condition, or a vertical datum. I therefore found **no basis for saying that the pole was definitely a SLOSH marker**.

SLOSH is nevertheless a plausible candidate for the planning data behind the concept. NHC's long-standing MEOW and MOM framework is explicitly designed for hurricane evacuation planning. A MEOW combines multiple hypothetical SLOSH tracks while holding category, forward speed, trajectory and initial tide assumptions constant; a MOM then takes maxima across MEOWs for a given hurricane category. In other words, **Category 4 and Category 5 worst-case planning surfaces are exactly the kind of product SLOSH can generate**. But that conceptual match is an inference, not documentary proof of this marker's source. ²⁵

NHC is emphatic about the interpretation: a MEOW is a worst-case envelope assembled from multiple hypothetical tracks, and **no single hurricane will produce the entire regional flooding pattern depicted by a MEOW**. Similarly, a MOM is a category-specific near-worst-case planning envelope rather than a forecast of one storm. ²⁶

That distinction may explain part of the Clear Lake reaction. A public marker has no map legend explaining ensemble maxima, track uncertainty, tide assumptions or return periods. It converts a probabilistic/planning abstraction into a giant physical object that looks like a simple promise: "A Category 5 puts the

water here.” Contemporary officials were simultaneously discovering that this category-to-surge shorthand was not scientifically sound. That interpretation is consistent with the sources but remains an inference.

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Why the Saffir-Simpson wording became obsolete

NHC's May 2009 notice is unusually direct. It proposed deleting storm-surge ranges and flooding references from all five categories because those surge ranges were often inconsistent with storms' wind categories and could mislead preparedness decisions; NHC specifically cited factors such as **storm size and landfall location**. The revised system became the **Saffir-Simpson Hurricane Wind Scale**, which today is explicitly based only on sustained wind speed. 28

That puts Sullivan's 2010–2011 statements in context. At the October town hall he said the sign language was “not appropriate any more,” and in the ABC13 interview he said the problem was the wording relative to the hurricane scales officials would use going forward—not simply whether residents liked the signs or whether the risk itself was right or wrong. 2

So the most defensible hackathon wording is:

The marker's extreme-surge hazard was not disproved. Its Category 4/Category 5 communication scheme became scientifically problematic because hurricane wind category does not uniquely determine storm surge. 28

Was 25 feet unrealistic?

Not inherently. ProPublica and the Texas Tribune's later Houston investigation examined two different extreme-surge models. Rice's SSPEED Center/University of Texas “**Mighty Ike**” or **P7+15** scenario modified an Ike-like storm and produced water exceeding 25 feet above sea level in portions of the Clear Lake/Galveston Bay system; a separate FEMA/USACE synthetic-storm analysis produced even higher levels in some locations. The article describes the scenarios as rare but physically meaningful planning cases rather than Category 5 hurricanes. 29

Those later calculations **must not be retroactively substituted for the marker's missing source model**. Their value is narrower: they show that the basic idea “Clear Lake can face an extreme 20–30-foot-class surge/water-level hazard under an unfavorable major hurricane” did not disappear when the marker was taken down. 30

Blackburn makes essentially that argument retrospectively. His 2023 Baker paper says the marker was not removed because the underlying danger had been shown to be wrong; he characterizes the warning as “right on target.” That is Blackburn's expert retrospective judgment, not evidence of the original TDEM engineering basis, and it should be presented as such. 13

The critical AR question: what does “25 feet” mean vertically?

This is the most important unresolved technical issue for your application.

There are at least three quantities that people casually call “a 25-foot surge”:

Storm-surge height relative to a reference water level. This describes the abnormal rise of coastal water generated by the storm.

Absolute water-surface elevation, for example 25 ft NAVD88 or another datum.

Inundation depth above local ground, which might be only several feet at a location whose terrain elevation is already well above the vertical datum.

The 2011 articles do not tell us which quantity was physically encoded on the Clear Lake marker. ²

That ambiguity is not academic. USGS explicitly notes that storm-surge observations from a large coastal region have to be brought to a **common national vertical datum**; a local datum is not adequate for combining measurements, and post-hurricane surveying uses GNSS/leveling to establish consistent water elevations. ³¹

Modern NHC products make the distinction much clearer. Operational Potential Storm Surge Flooding products present inundation **above ground level**, while probabilistic guidance has also been provided relative to **NAVD88**. NHC uses high-resolution elevation data to turn water-level modeling into maps of flooding over normally dry ground. ³²

That leads to a practical AR rule:

Do not use “25 ft” from the 2011 newspaper as an absolute AR ^z coordinate.

For a historically faithful **visual replica of the pole**, a provisional ~25-ft object is defensible if you explicitly label its dimensions approximate. For a scientifically meaningful **water surface**, use either:

water-surface elevation in a known datum - terrain elevation in the same datum =
depth above ground

or use an authoritative dataset that already gives **depth above ground**. USGS/NHC guidance demonstrates why mixing vertical reference systems without this transformation can produce misleading results. ³³

Modern NHC SLOSH-based national risk maps are also useful as context but not as parcel-level truth: NHC creates category-specific MOM inundation products, using high-tide scenarios and one-foot depth bins, while explicitly warning that the data are for community-level planning/awareness rather than parcel-level determinations. ³⁴

Controversy, removal, and the Blackburn connection

The controversy had **two overlapping strands** that are sometimes collapsed into one story.

The first was economic and psychological. Contemporary reporting says some residents and real-estate agents feared the markers could depress property values. ABC13 interviewed resident Susan Maki, who said she simply did not believe water would get that high, and reported that local real estate professionals were

fielding worried calls from prospective or current residents at a time of economic anxiety associated with NASA layoffs. ³⁵

Todd Hinz, identified as a Bay Oaks resident, lobbied Houston Councilmember Mike Sullivan for months. Hinz's objection was partly comparative: Ike had not produced anything resembling the marker's displayed depth in Clear Lake, whereas places such as Kemah and Seabrook that had experienced dramatic Ike impacts did not have such signs. ABC13 quotes him arguing that outsiders might see the markers and conclude that something was wrong with Clear Lake. ¹⁸

The second strand was legitimate risk-communication science. Sullivan's explanation was not "the rich homeowners made me do it." He told ABC13 that the issue was not simply whether the signs were right or wrong or liked or disliked, but that their **language was inappropriate for the scales officials would use for future hurricanes**. That explanation is highly consistent with NHC's contemporaneous removal of surge ranges from the Saffir-Simpson scale. ³⁶

Mayor Annise Parker had already halted the rest of the six planned Houston/Clear Lake installations in 2010. Her letter, as quoted contemporaneously, said the city was **"reassessing the markers and their locations due to possible misinterpretation."** ¹⁰

Around mid-January 2011, **City of Houston workers** removed the two markers identified by the newspaper. The evidence I found does not identify a formal Houston City Council vote, a written mayoral removal order, or an instruction from FEMA directing Houston to remove them. ABC says that after months of lobbying Sullivan "the decision was made" to pull them; the *Bay Area Citizen* names Houston crews as the people who physically did so. Therefore the strongest conclusion is **local Houston termination/removal after community and political pressure**, not "FEMA was forced to remove its pole." ²

Similarly, I found evidence of concerns voiced by residents and people in real estate, but **not evidence that a Realtor association or organized real-estate lobby formally ordered or legally compelled removal**. That distinction is worth preserving in a hackathon presentation. ²

At the broader program level, Texas DPS spokesperson Tom Vinger later told ProPublica/Texas Tribune that the marker program was **"discontinued due to lack of community interest."** That suggests the Clear Lake backlash was part of a broader difficulty obtaining local acceptance, although the surviving reporting does not establish that Houston alone caused the statewide/regional program to end. ¹⁹

Blackburn and the photograph you saw

Your recollection that **Jim Blackburn was using this marker story in roughly 2018 is strongly corroborated**.

On November 12, 2018, Blackburn published *The Houston Plan for Flood Damage Reduction* through Rice's Baker Institute. In the section on public information, he wrote that a Clear Lake sign showing Category 4/5 surge levels had been removed because it interfered with home sales and argued that Houston needed to be more candid about genuine flood risk. ²⁰

2018 Baker brief:

<https://www.bakerinstitute.org/research/houston-plan-flood-damage-reduction>

I did not recover the specific 2018 Ike-Dike presentation or slide deck you remember. There were multiple Houston coastal-protection discussions involving Blackburn in that period; for example, November 2018 reporting records him publicly criticizing aspects of the Corps' coastal-spine analysis and arguing that the Clear Lake/Ship Channel region retained substantial surge exposure. That establishes the right context, though not the specific talk. ³⁷

By 2023, Blackburn and Jennifer Borski made the anecdote much more explicit in the Baker Institute's Harvey retrospective. They published **Figure 3**, captioned as a Clear Lake-area flood marker showing Category 4 and Category 5 surge levels, and the caption identifies Blackburn as the photographer. The accompanying text says the marker was removed following complaints that it interfered with home sales. ¹³

That makes Blackburn the single most promising human source for the AR reconstruction. He may possess the **original photograph rather than the web-sized reproduction, additional frames from the same visit, the date/location metadata, and the 2018 presentation file you actually saw**. Those items could resolve the profile, height, top-panel wording and precise installation site far faster than another round of generic web searching.

There is also a striking conceptual through-line for your hackathon project. The physical marker was an attempt to turn a low-frequency, abstract hazard into something visible at human scale. Blackburn's 2018 writing argues that Houston suffers when flood information is hidden or softened; modern NHC products increasingly express surge as **depth above ground** for exactly the reason that people need to understand what the water means where they stand. Your AR project recreates the perceptual function of the physical marker while allowing the underlying scenario, probability, event, datum and uncertainty to be explicit and changeable. ³⁸

Unresolved records, archival targets, and the most useful AR findings

The major unanswered questions are no longer vague. They are quite specific, which is good news for subsequent archival work.

What is still genuinely unknown

Question	Status after this research	Best record likely to resolve it
Official name of the marker program	Unknown	TDEM/DPS project file; FEMA Region 6 correspondence
FEMA grant/program and award number	Unknown	FEMA/TDEM grant ledger; post-Ike mitigation records
Exact federal/state cost split	Unknown	Grant award/budget
Exact number fabricated	"Hundreds" only. ¹⁹	Vendor invoice/purchase order

Question	Status after this research	Best record likely to resolve it
Exact number installed on upper Texas coast	Unknown	TDEM site inventory/ completion report
Exact number installed in Clear Lake	At least two; potentially “several” depending on ABC wording. ²	Houston work orders/site list
Identities of four other planned Clear Lake locations	Unknown	Parker correspondence / site plan
Exact lat/lon of the two known poles	Unknown	Installation work orders, photos, GIS or Street View
Exact pole height and width	Unknown; ~25 ft is plausible but not specification-grade. ¹²	Fabrication/shop drawing
Exact colored-band elevations	Unknown	Artwork/design drawing
Top placard text	Unreadable in surviving public image	Original Blackburn photo / ABC footage
Material, support and footing	Unknown	Fabricator drawing/work order
Manufacturer/fabricator	Unknown	Purchase order/invoice
Designer	Unknown	Project correspondence
Source model	Unknown; SLOSH/MOM plausible, unproven	TDEM technical memo/model map
Vertical datum	Unknown	Technical design document
Were marker values AGL, surge height, or absolute elevation?	Unresolved	Same technical document
Exact 2018 Blackburn slide deck	Not recovered	Blackburn/SSPEED/Rice files

Records requests with the highest expected payoff

A **TDEM/Texas DPS public-records request** should ask for every record from roughly 2009–2011 containing “storm surge marker,” “surge marker,” “Clear Lake,” and the \$56,000 project amount, including contracts, purchase orders, invoices, artwork, engineering specifications, grant numbers, FEMA correspondence, site lists and project-closeout documentation. The present TDEM Hazard Mitigation Section is the logical institutional starting point, although the project may have originated elsewhere in the older DPS-era TDEM organization. ³⁹

TDEM mitigation page:

<https://www.tdem.texas.gov/mitigation/hazard-mitigation-section>

A **City of Houston records request** should target 2010–January 2011 files of Mayor Annise Parker and then-District E Councilmember Mike Sullivan using the names **Todd Hinz, storm surge marker, Clear Lake City Boulevard, Space Center Boulevard, Bay Area Boulevard, and Clear Lake High School**. Particularly valuable records would be Parker's quoted letter, incoming correspondence from residents/real-estate interests, Public Works removal work orders, and any site drawings or utility-clearance documentation. The contemporary newspaper proves that such correspondence and city removal activity existed. ¹⁰

A **Rice/SSPEED/Blackburn inquiry** has arguably the best immediate odds of helping tomorrow's 3-D model: request the original-resolution file behind Baker Figure 3, any adjacent photographs from the same roll/visit, its EXIF/date/location if preserved, and the slide deck from Blackburn's 2017–2019 coastal-protection/flood talks containing the photograph. The published Baker caption establishes that Blackburn himself supplied the image. ¹³

An **ABC13/KTRK archive request** for the February 9, 2011 package could be even more valuable geometrically. Television footage often includes pans from top to bottom, interviews with people standing near the object, cars, road signs and known-height infrastructure. Any of those would let you solve approximate dimensions photogrammetrically. The surviving ABC article gives the exact date and story title needed for an archive librarian to find it. ¹⁸

A **Houston Chronicle/Bay Area Citizen photo-archive request** for Mary Ann Hellinghausen's February 1, 2011 story should request all assigned and unpublished images, not merely the image published online. The current article retains the contemporaneous story but does not expose a high-resolution source image in the searchable material I recovered. ¹⁰

A **FEMA Region 6 / FEMA records request** should avoid presupposing that the project was HMGP. Ask instead for records relating to the "Texas upper coast storm surge marker public-awareness project," approximately \$56,000, circa 2009–2011, and request the funding program, award/project number, scope of work, state subaward, technical basis and design documents. That formulation lets FEMA identify the funding stream rather than forcing an unverified HMGP assumption.

What is most useful for the AR model right now

For tomorrow's hackathon, there is enough evidence to build two deliberately different versions.

The first can be a **historical reconstruction**, clearly labeled approximate. Use the Blackburn silhouette, blue lower body, green upper body, prominent vertical Category 4/5 lettering, and a provisional overall scale in the **20–25-foot range**. Do not invent detailed tick marks, manufacturer logos or top-panel wording. The correct claim is "reconstruction from surviving photograph and contemporary descriptions," not "digitally exact replica." ⁴⁰

The second should be the actual **AR flood-information concept**, which fixes the old marker's principal communication weakness. Rather than saying merely "Category 5," the AR overlay can identify the scenario:

Historical Clear Lake FEMA/Texas marker — reconstructed

Extreme surge planning scenario, circa 2010

Original model/datum still under archival investigation

and for modern data:

Modeled inundation: 12 ft above ground

Source: NHC SLOSH/P-Surge

Scenario/probability: displayed explicitly

Terrain reference: current geospatial ground surface

That design follows modern NHC practice of separating wind category from storm-surge risk and presenting operational inundation in terms of **water above ground**, while preserving the extraordinary communicative insight of the old physical pole: let a person look up and comprehend where the water actually is. ⁴¹

For the hackathon narrative, this is a source-supported formulation:

After Hurricane Ike, FEMA and Texas funded a storm-surge-awareness project that placed unusually tall Category 4/5 surge markers in Clear Lake. Six were planned in Houston; contemporary reporting identifies at least two installations. Residents and real-estate interests objected to the markers' effect on perceptions of the community, while officials also recognized that category-based surge wording no longer reflected emerging National Hurricane Center guidance. Houston halted further installation and removed the two documented signs in early 2011. Rice University flood expert Jim Blackburn continued using one of the markers as an example of the difficulty of communicating Houston's flood risk and preserved a photograph of it. Our AR project takes the original idea—making invisible risk visible at human scale—and puts the marker back virtually, with modern geospatial data and explicit scenario, probability and vertical-reference information. ⁴²

That last sentence is where your project's history and technology meet. The most compelling version is not "FEMA put up a scary pole and Realtors made them remove it." The evidence supports a better story: **officials tried to make an extreme but legitimate hazard physically comprehensible; the visualization was politically explosive and scientifically oversimplified; it disappeared; and AR now makes it possible to restore the visualization while improving the science behind what it communicates.**

Key primary and near-primary research links

The most useful documents/pages recovered in this pass are:

Contemporary Clear Lake removal report — *Bay Area Citizen/Houston Chronicle*, February 1, 2011:

<https://www.chron.com/neighborhood/bayarea/news/article/Storm-surge-markers-removed-9299639.php>

¹⁰

Contemporary ABC13 report, February 9, 2011:

<https://abc13.com/archive/7946821/> ¹⁸

Jim Blackburn & Jennifer Borski, Baker Institute, 2023 — includes authenticated marker photograph:

<https://www.bakerinstitute.org/research/assessing-houstons-flood-vulnerability-6-years-after-harvey> ¹³

Direct Blackburn marker photograph:

<https://www.bakerinstitute.org/sites/default/files/2023-08/Blackburn-6years-Harvey-080423-Fig03.png> 13

Blackburn, Baker Institute, November 2018 — marker/home-sales anecdote near the time you remember hearing him:

<https://www.bakerinstitute.org/research/houston-plan-flood-damage-reduction> 20

ProPublica/Texas Tribune, 2016 — broader marker program and extreme-surge investigation:

<https://www.propublica.org/article/hell-and-high-water-text> 19

Associated interactive/model material:

<https://projects.propublica.org/houston/> 43

NHC 2009 notice removing surge ranges from the Saffir-Simpson categories:

https://www.nhc.noaa.gov/news/pns10_2009.shtml 17

NHC SLOSH MEOW technical description:

<https://www.nhc.noaa.gov/surge/meowDescrip.php> 44

NHC SLOSH MOM technical description:

<https://www.nhc.noaa.gov/surge/momDescrip.php> 45

NHC National Storm Surge Risk Maps:

<https://www.nhc.noaa.gov/nationalsurge/> 34

NHC Potential Storm Surge Flooding Map methodology:

<https://www.nhc.noaa.gov/surge/inundation/> 46

USGS explanation of common vertical datums in storm-surge surveying:

<https://water.usgs.gov/osw/gps/storm-surge.html> 31

TDEM Hazard Mitigation Section:

<https://www.tdem.texas.gov/mitigation/hazard-mitigation-section> 39

The principal archival gap remains the **original FEMA/TDEM project folder**. Recovering it—or Blackburn's original photo set or the ABC13 footage—would likely produce a disproportionate payoff: the grant identity, complete site list, fabrication/vendor information, model basis, and especially the dimensional evidence needed to turn the provisional AR marker into a historically accurate 3-D reconstruction.

1 2 4 6 7 8 10 11 22 23 24 27 42 Storm surge markers removed

https://www.chron.com/neighborhood/bayarea/news/article/Storm-surge-markers-removed-9299639.php?utm_source=chatgpt.com

3 17 28 nhc.noaa.gov

https://www.nhc.noaa.gov/news/pns10_2009.shtml?utm_source=chatgpt.com

- 5 13 21 40 Assessing Houston's Flood Vulnerability 6 Years After Harvey | Baker Institute
https://www.bakerinstitute.org/research/assessing-houstons-flood-vulnerability-6-years-after-harvey?utm_source=chatgpt.com
- 9 16 19 29 30 Hell and High Water (Full Text) — ProPublica
https://www.propublica.org/article/hell-and-high-water-text?utm_source=chatgpt.com
- 12 How one artist gets homeowners to advertise their homes' flood risk - The Washington Post
https://www.washingtonpost.com/climate-solutions/2023/12/13/xavier-cortada-the-underwater/?utm_source=chatgpt.com
- 14 nhc.noaa.gov
https://www.nhc.noaa.gov/archive/2008/al09/al092008.public.049.shtml?utm_source=chatgpt.com
- 15 Hazard Mitigation Assistance | FEMA.gov
https://www.fema.gov/to/node/623465?utm_source=chatgpt.com
- 18 35 36 Clear Lake removes storm surge warning signs after residents complain of negative impact on home values | ABC7 Los Angeles | abc7.com - ABC7 Los Angeles
https://abc7.com/archive/7946821/?utm_source=chatgpt.com
- 20 38 The Houston Plan for Flood Damage Reduction | Baker Institute
https://www.bakerinstitute.org/research/houston-plan-flood-damage-reduction?utm_source=chatgpt.com
- 25 44 Storm Surge Maximum Envelope of Water (MEOW)
https://www.nhc.noaa.gov/surge/meowDescrip.php?utm_source=chatgpt.com
- 26 Storm Surge Maximum Envelope of Water (MEOW)
https://www.nhc.noaa.gov/surge/meowOverview.php?utm_source=chatgpt.com
- 31 33 U.S. Geological Survey - Global Positioning System
https://water.usgs.gov/osw/gps/storm-surge.html?utm_source=chatgpt.com
- 32 nhc.noaa.gov
https://www.nhc.noaa.gov/templates/graphics_esurge.shtml?utm_source=chatgpt.com
- 34 National Storm Surge Risk Maps - Version 4
https://www.nhc.noaa.gov/nationalsurge/?utm_source=chatgpt.com
- 37 Ike Dike concept raises more questions than it answers for environmentalists and Rice University
https://www.houstonchronicle.com/news/houston-texas/houston/article/Ike-Dike-concept-raises-more-questions-than-it-13378121.php?utm_source=chatgpt.com
- 39 Hazard Mitigation Section
https://www.tdem.texas.gov/mitigation/hazard-mitigation-section?utm_source=chatgpt.com
- 41 Saffir-Simpson Hurricane Wind Scale
https://www.nhc.noaa.gov/aboutsshws.php?utm_source=chatgpt.com
- 43 Hell and High Water: Why Isn't Texas Ready for the Next Big Hurricane? | ProPublica and The Texas Tribune
https://projects.propublica.org/houston/?utm_source=chatgpt.com
- 45 Storm Surge Maximum of the Maximum (MOM)
https://www.nhc.noaa.gov/surge/momDescrip.php?utm_source=chatgpt.com
- 46 Potential Storm Surge Flooding Map
https://www.nhc.noaa.gov/surge/inundation/?utm_source=chatgpt.com